

Seat No: _____

Enrollment No: _____

PARUL UNIVERSITY
FACULTY OF IT & COMPUTER SCIENCE
MCA/M.Sc.IT 2025-26 Mid-Term Examination

Semester: 3

Date: - -2025

Subject Code: 05201352

Time: (1hr:30min)

Subject Name: Cyber Security & Forensics - II

Total Marks: 40

Instructions:

1. Figures to the right indicate full marks.
2. Make suitable assumptions wherever necessary.

Q.1	Answer the following.	CO	PO	BL	[10]
(a)	3 short questions of 1 mark each				[3]
	(i) Define CIA triad in cybersecurity	1	1	Remembering (BT-1)	
	(ii) What is the purpose of virtualization?	1	1	Remembering (BT-1)	
	(iii) Give one example of symmetric encryption.	3	3	Remembering (BT-1)	
(b)	Objective type/MCQs/True-False/Fill in blanks (7 questions of 1 mark each)				[7]
	1. Which tool is commonly used for port scanning in Kali Linux? a) Wireshark b) Nmap c) Hydra d) Metasploit	1	1	Remembering (BT-1)	
	2. True or False: with justification Steganography hides information in encrypted files.	3	3	Understanding (BT-2)	
	3. Which port number is used by HTTPS? a) 21 b) 22 c) 443 d) 80	1	1	Remembering (BT-1)	
	4. Fill in the blanks _____ is used to convert plaintext into ciphertext.	2	2	Remembering (BT-1)	
	5. Which of these is a feature of a firewall? a) Encrypt data b) Filter traffic c) Store passwords d) Scan QR codes	1	1	Understanding (BT-2)	
	6. What does IDS stand for in cybersecurity? a) Internal Digital Shield b) Internet Detection System c) Intrusion Detection System d) Inbound Data System	1	1	Remembering (BT-1)	
	7. What is the use of hashing? a) Encrypt data b) Store passwords securely c) Open ports d) Configure firewalls	4	4	Understanding (BT-2)	
Q.2	Answer the following. (2 or 3 mark questions)				[10]
(a)	Two Questions of 2 Marks				[4]
	(i) What is a vulnerability? Mention one example.	1	1	Understanding (BT-2)	(02)
	(ii) Describe any two types of cyber threat actors.	1	1	Understanding (BT-2)	(02)
(b)	Two Questions of 3 Marks				[6]
	(i) Explain how firewalls help prevent unauthorized access.	1	1	Understand (BT-2)	(03)
	(ii) Describe the role of Burp Suite in web application testing.	4	4	Apply (BT-3)	(03)
Q.3	Attempt any TWO.				[10]
	(i) Describe different types of firewalls with examples.	1	1	Apply (BT-3)	(05)
	(ii) Analyze the working of digital signatures in ensuring integrity.	4	4	Analyse (BT-4)	(05)
	(iii) Evaluate the importance of secure	1	1	Evaluate (BT-5)	(05)

	communication protocols like SSH and TLS.				[10]
Q.4	Answer the following.	1	1	Apply/Analyze (BT-3/4)	(05)
(a)	Use Case: A university's IT department notices that large files transferred between campuses are either getting lost or arriving corrupted. Upon inspecting packet captures, you find that some packets have very low Time to Live (TTL) values and incorrect Fragment Offset entries. Question: Analyze how the TTL and Fragment Offset fields work in the IP header to support packet delivery. What issues might occur if these fields are misconfigured or manipulated, and how could it affect network communication?				
(b)	Evaluate the use of TCP versus UDP in different network services like file transfer, video streaming, and DNS resolution. Which protocol would you recommend for each case and why? OR An organization is planning to use either static routing or dynamic routing protocols. Evaluate both approaches and recommend which one would be more efficient for a medium-sized, growing network. Justify your answer.	1	1	Evaluate	(05)
(b)		1	1	Evaluate (BT-4/5/6)	(05)